

STRUCTURAL BIOINFORMATICS REPORT

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1. Introduction

The aim of this project is to structurally and functionally characterize the provided unknown protein sequences using structural bioinformatics approaches. If it interacts with other nucleic acids, proteins, or ligands. Additionally, the work also aims for the functional characterization of the given unknown proteins, which will help to provide the information about the protein if it belongs to protein family and functionally important residues such as active sites, binding sites and its possible role in the future experiments. The overall objective is to infer the possible biological roles of the proteins and provide comprehensive structural and functional annotations supported by computational analyses and structural models.

FASTA Sequences Provided-

>Baltymas_1

MHLISFLGTGNYSETTYRFDTQTCRTRYVALALAQFTRPDKVRILATDEAWQQHGQ
GLSHALTDAGFAEPERLAIPTGGEPEQLWKMFEAIVDAIASAGESVMLDITHGFRMQ
PFFAAACIEYVQSVLKNRPSICVVYGEFRKDAPESPIWELTPFLEVLEWSRNLMVFLR
TGQAAAVVNSTEALARRLNRQWAESGREGPQPKLMKLARSMERFSNDFCTVRTGS
LLVGSNSSVAELLQAIQVTEDEVRQRLPVLAPVLKELRDVVQPLEIGGERFSSRRGQE
VLLALARLYESMGRFSEAYAILREGWISLGAPPEADTPNTPEFDFKARDRWDKAWP
RRAGYPNQVSEPRNDLLHAGFRDQPLPPKSITEQLSSLLTQWEDEIRKPSNANINKVG
GLMIILNFGHTITDRHRQEIEALTGQKVHDVLAIPTQFDHSRSFSDQIIDLLQTIPFDAE
AWQSKSILVNPPSFAAITAVLMAELHGRCGYFPPIIRFRPIPNLVPPQFEVAEIINLQEIR
DAARKRRG

>Baltymas_2

MDIASIKQAIRQPSVQADKLPVHIDACCRFLSPWTKPSCEIQGLESDGSYRRIAAIGFA
LAQNRQSETLQKAFVEGTQQLAGRVYFTLGRAPTVEVDGVALLGLAIGYNALTPST
MDDGWLRVCLVQSVAALQNDPWQYSLAKSAQATLGVSIDKSDIDPVLRVAVAERK
GDVIEEELRGAAWCLLLENIDEPEPTRRAALQGTFESCASALARLPLHGAGVSELIEIL
EGVSRSMGHWTYEEKPKVRNVLPQKWEIDHEYHVQNLLWTILRPIFPDLVDEETLK
KLGHTSPRYDLGIPSLSTIIEVKYVRRRGQSALKAITDEIAADHSLYLREGTGFARMIA
FIWDEQRQTEEYQTLQSGLENLSGLERVIIVPRPAKMERDKKK

2. Methodology

To learn about the unknown protein, a simple sequence- based analysis was performed:

2.1 Sequence Similarity Searches

The analysis began with identifying the homologs for both the protein sequences; BLASTp (Basic Local Alignment Search Tool protein) [1] was run against the NCBI non-redundant (NR) protein database. To detect distant evolutionary relationships, position-specific iterative BLAST (PSI-BLAST) [2] was performed for up to three iterations against NR and uniRef50 [3] hits were collected below e-value threshold 1e-3.

For sensitive homology searches of the sequences HHpred tool [5] was used. And also, to gain a better understanding of the domain architecture of protein sequences. Both unknown protein sequences were searched against multiple databases including: **PDB_mmCIF70_20Feb, ECOD_F70_20230309, CATH_S40_v4.3, and Pfam v38.2.**

HHpred | Bioinformatics Toolkit

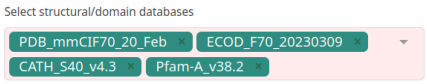


Figure 1 HHpred domain databases

2.2 3D Structure Prediction

In order to understand protein functions,the 3D structure of both proteins was generated with the help of AlphaFold 3 [6]. Models quality was assessed with iPTM and pTM scores and most suitable model was selected for further studies. Whereas the predicted aligned error (PAE matrices) shows the confidence of interdomain relationships. https://alphafoldserver.com/

Furthermore, the quality of the predictied structures of both proteins, VoroMQA has been used. This evaluates and produces scores at atomic, residue, and global levels, all in the fixed range from 0 to 1. [7].

https://bioinformatics.lt/wtsam/voromqa

The structural models generated from alphafold 3 and VoroMQA were visualized and analyzed using PyMOL [8].

2.3 Structural domain annotations and Fold Classification

The query protein domain boundary and its architecture was estimated by CHAINSAW [9] followed by Alphafold 3 and VoroMQA.

To predict domain structures of the query protein, on the basis of their tertiary structure and to study about their evolutionary relationships- DALI (Distance matrix alignment) tool [10] was used. It generates structural alignments and measures the degree of similarity by generating Z-scores, RMSD values and sequence identity of the similar proteins. The resulting structural models were inspected and visualized using PyMOL.

Dali server

2.4 Prediction of conserved sites

To identify the evolutionary conserved amino acids in both the proteins, ConSurf analysis was carried out. Highly conserved regions in the protein are often indicates the structural stability, catalytic activity, ligand binding, or protein-protein interactions [12] ConSurf | Evolutionary conservation profiles of proteins

2.5 Putative binding sites and heteromeric interaction

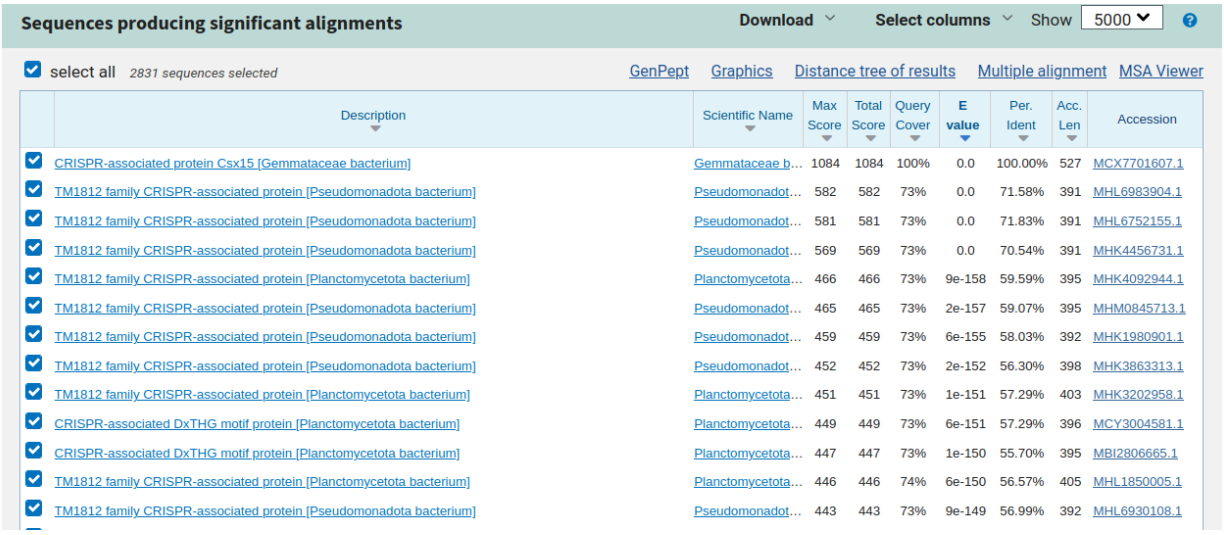
Putative binding sites were predicted using the structure and sequence based tools to determine protein ligand or protein-protein interacting residues. Heteromeric interactions were evaluated through protein modeling and interface analysis.

3 Results Of Baltymas 1

3.1 Sequence analysis:

To begin with the search of homologs of baltymas 1. The BLASTp results indicate that Baltymas 1 is a homolog to the CRISPR-associated protein csx15 from a Gemmataceae bacterium (figure 2). Due to these reasons:-

- 100% query coverage,
- E-Value = 0.0,
- 100% Identity



Sequences producing significant alignments Download Select columns Show 5000

select all *2831 sequences selected* GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
CRISPR-associated protein Csx15 [Gemmataceae bacterium]	Gemmataceae b...	1084	1084	100%	0.0	100.00%	527	MCX7701607.1
TM1812 family CRISPR-associated protein [Pseudomonadota bacterium]	Pseudomonadot...	582	582	73%	0.0	71.58%	391	MHL6983904.1
TM1812 family CRISPR-associated protein [Pseudomonadota bacterium]	Pseudomonadot...	581	581	73%	0.0	71.83%	391	MHL6752155.1
TM1812 family CRISPR-associated protein [Pseudomonadota bacterium]	Pseudomonadot...	569	569	73%	0.0	70.54%	391	MHK4456731.1
TM1812 family CRISPR-associated protein [Planctomycetota bacterium]	Planctomycetota...	466	466	73%	9e-158	59.59%	395	MHK4092944.1
TM1812 family CRISPR-associated protein [Pseudomonadota bacterium]	Pseudomonadot...	465	465	73%	2e-157	59.07%	395	MHM0845713.1
TM1812 family CRISPR-associated protein [Pseudomonadota bacterium]	Pseudomonadot...	459	459	73%	6e-155	58.03%	392	MHK1980901.1
TM1812 family CRISPR-associated protein [Pseudomonadota bacterium]	Pseudomonadot...	452	452	73%	2e-152	56.30%	398	MHK3863313.1
TM1812 family CRISPR-associated protein [Planctomycetota bacterium]	Planctomycetota...	451	451	73%	1e-151	57.29%	403	MHK3202958.1
CRISPR-associated DxTHG motif protein [Planctomycetota bacterium]	Planctomycetota...	449	449	73%	6e-151	57.29%	396	MCY3004581.1
CRISPR-associated DxTHG motif protein [Planctomycetota bacterium]	Planctomycetota...	447	447	73%	1e-150	55.70%	395	MBI2806665.1
TM1812 family CRISPR-associated protein [Planctomycetota bacterium]	Planctomycetota...	446	446	74%	6e-150	56.57%	405	MHL1850005.1
TM1812 family CRISPR-associated protein [Pseudomonadota bacterium]	Pseudomonadot...	443	443	73%	9e-149	56.99%	392	MHL6930108.1

Figure 2 BLASTp search against non-redundant databases from NCBI for Baltymas_1

Supplementary:

Structural_bioinformatics_files > Supplementary > Baltymas1_fasta_files

To make stronger functional assessment HHpred tool was used. Where, each bar represents homologous protein and the length of the bar indicates how much proteins aligned with the

query protein. The first hit was 'TIGR02221 family CRISPR-associated proteins' (figure 3) that indicates that the query protein is closely related to CRISPR-associated proteins family.

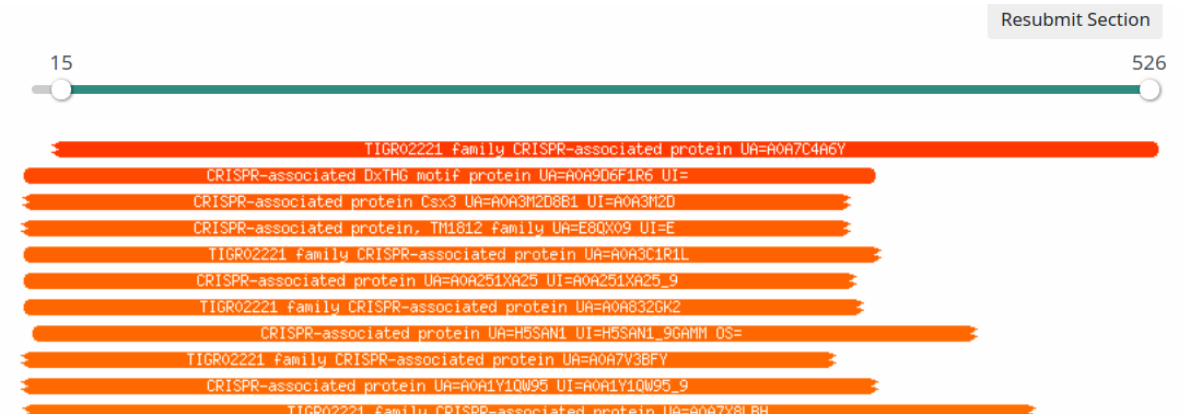


Figure 3 HHpred search result of baltymas 1

3.2 Structure prediction:

The 3D structure of the baltymas 1 sequence was analyzed and modeled by using Alphafold3 server as shown in (Figure 4). The iPTM scores are 0.71 that indicated very good confidence in the intermolecular interface. The 3D structure shows the pLDDT higher values and few terminals displayed the lower confidence scores. This suggests that the overall predicted structure closely resembles its true confirmation.

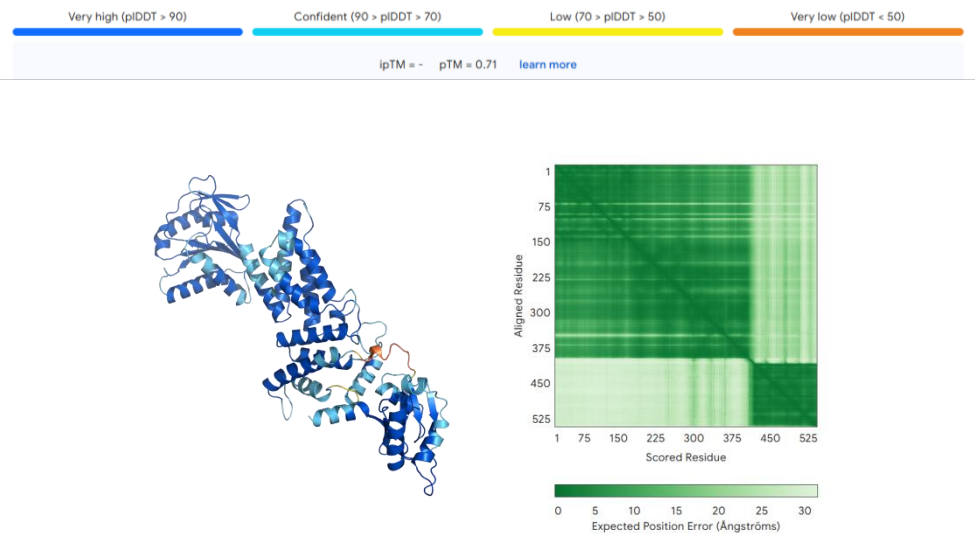


Figure 4 Baltymas 1 prediction from Alphafold 3 server (https://alphafoldserver.com/) and the 3D structure is visulaised by PyMOL.

Supplementary: Structural_bioinformatics_files > Supplementary > Alphafold

The structure quality analysis of baltymas 1 was done by VoroMQA. The global scores from VoroMQA were reported to be 0.519, indicating overall good quality as shown in figure 5.

This further validates the structural model as a suitable candidate for downstream analyses, including functional characterization.

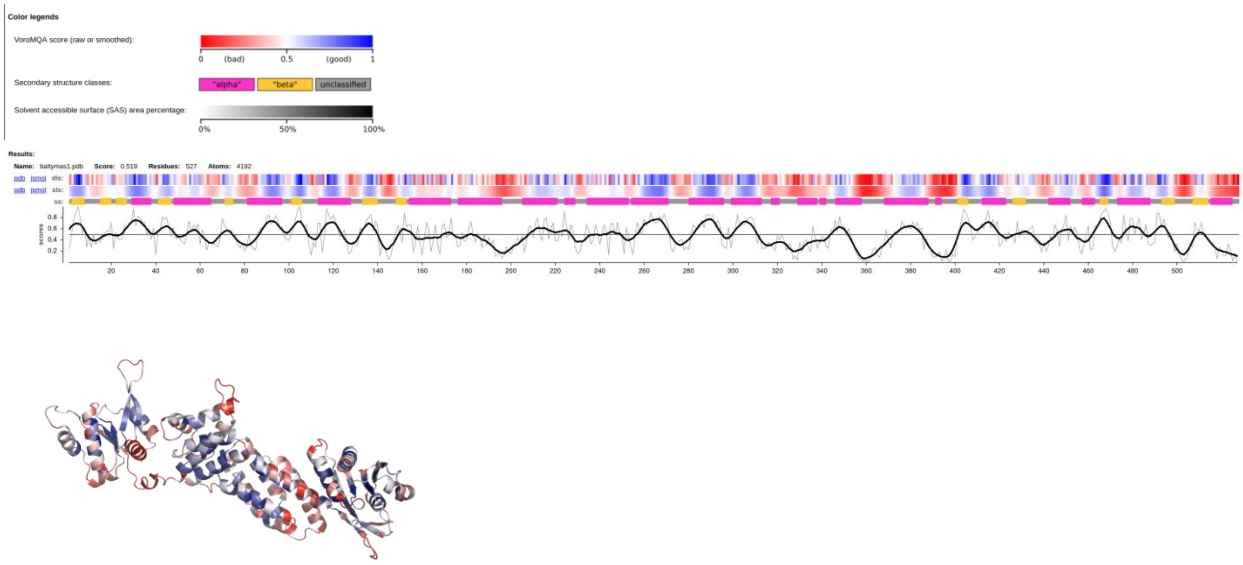


Figure 5 VoroMQA analysis of Baltymas 1 for structure quality analysis. The generated structure visualised with the help of PyMOL

3.3 **Domain segmentation**

Following the structure prediction of Baltymas 1 using AlphaFold3, a structural similarity search was performed using the DALI server to identify the closest structural homologues in the PDB as shown in figure 6. The data revealed a strong alignment between the query structure and the 8y7g-B with a z-score of 22.1 which is well above the significance threshold. The RMSD value is 3.9 Å was obtained based on 543 aligned residues.

No:	Chain	Z	rmsd	lali	nres	%id	PDB	Description
1:	8y7g-B	22.1	3.9	387	543	20	PDB	MOLECULE: CRISPR-ASSOCIATED PROTEIN;
2:	6qzq-A	20.3	4.3	349	454	18	PDB	MOLECULE: CRISPR-ASSOCIATED (CAS) DXTHG FAMILY;
3:	9teu-A	19.2	1.7	129	131	33	PDB	MOLECULE: CSX15;
4:	9tet-A	14.6	2.1	117	127	25	PDB	MOLECULE: CRISPR-ASSOCIATED CSX15;
5:	9ylu-A	13.6	7.5	144	295	33	PDB	MOLECULE: CBCASH;
6:	1xmx-A	13.1	18.6	136	380	15	PDB	MOLECULE: HYPOTHETICAL PROTEIN VC1899;
7:	7bdv-A	12.7	18.5	132	354	12	PDB	MOLECULE: CAN2;
8:	2wte-A	12.4	2.9	156	212	14	PDB	MOLECULE: CSA3;
9:	8bao-A	12.2	17.7	135	368	11	PDB	MOLECULE: DUF1887 FAMILY PROTEIN;
10:	4eog-A	12.1	4.5	176	466	16	PDB	MOLECULE: PUTATIVE UNCHARACTERIZED PROTEIN;
11:	6tug-B	11.9	5.8	219	415	10	PDB	MOLECULE: CRISPR SYSTEM ENDORIBONUCLEASE CSM6;
12:	8jbc-A	11.8	4.0	204	468	16	PDB	MOLECULE: CRISPR SYSTEM ENDORIBONUCLEASE CSM6;
13:	2i71-B	11.7	7.2	177	376	15	PDB	MOLECULE: HYPOTHETICAL PROTEIN;
14:	8pe3-A	11.5	5.7	197	386	12	PDB	MOLECULE: CRISPR SYSTEM ENDORIBONUCLEASE CSM6';
15:	6o6s-A	11.5	7.3	189	433	16	PDB	MOLECULE: CSM6;
16:	7ygh-B	11.3	2.3	125	184	18	PDB	MOLECULE: CRISPR SYSTEM RING NUCLEASE SSO2081;
17:	8z3k-B	11.3	9.9	175	615	10	PDB	MOLECULE: ADENOSINE DEAMINASE DOMAIN-CONTAINING PROTEIN;
18:	6x11-A	11.2	19.4	130	373	15	PDB	MOLECULE: CAD1;
19:	8pcw-A	11.2	4.4	217	352	11	PDB	MOLECULE: CRISPR-ASSOCIATED PROTEIN CSM6;
20:	9u49-A	11.1	19.3	139	399	16	PDB	MOLECULE: CARF-DOMAIN CONTAINING DUAL-ACTIVATED NUCLEASE;
21:	9c67-B	11.1	8.4	158	540	8	PDB	MOLECULE: ADENOSINE DEAMINASE DOMAIN-CONTAINING PROTEIN;

Figure 6 DALI results for Baltymas_1

For more detailed structural comparison, DALI's first hit that was 8y7g – B (CRISPR – ASSOCIATED PROTEIN) was super imposed with baltymas 1 as shown in (figure 7), gave the RMSD value of super imposition is 3.538 (1802 to 1802 atoms). The structure superimposition of Baltymas 1 (shown in yellow) with its closest DALI hit csx1-crn2, represents the structurally conserved and aligned portion of **csx-1**. Notably, a distinct red region corresponding to a segment of csx-1 that could not superimpose with Baltymas 1, indicates that it may possess a unique structural architecture not fully represented by any currently known protein structure in PDB.

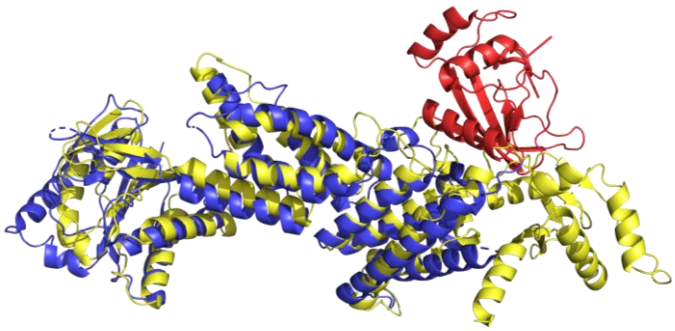


Figure 7 This is the visualization from PyMOL which shows the super impose of baltymas 1 (Yellow), DALI's first hit with blue colour indicates the superimposition of baltymas_1 and red colour of the DALI's first hit indicated the no imposition of a part of Baltymas 1

Supplementary: Structural_bioinformatics_files > Supplementary > DALI > baltymas1_csx1_active_site_pymol

To determine the function of the uncharacterised domain of the baltymas 1, structural superimposition with DALI hits was conducted. Superimposition with the third DALI hit 9teu (csx-15) confirmed that unpredicted domain of Baltymas 1 corresponds to Csx–15 with RMSD 1.767385 over 128 residues (shown in figure 8).

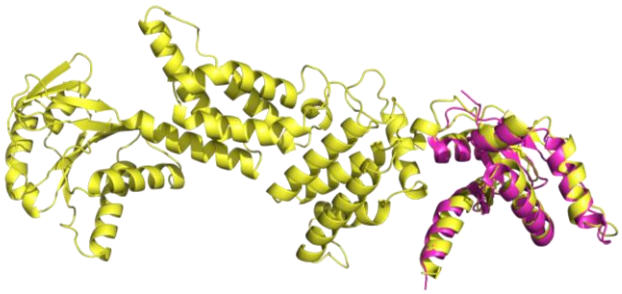


Figure 8 This is the visulization from PyMOL which shows the super impose of baltymas 1 (Yellow), DALI's hit (pink) that is csx-15, 3-D structure. Observed with the help of PyMOL.

Supplementary:

Structural_bioinformatics_files > Supplementary > DALI > baltymas1_csx1_active_site_pymol

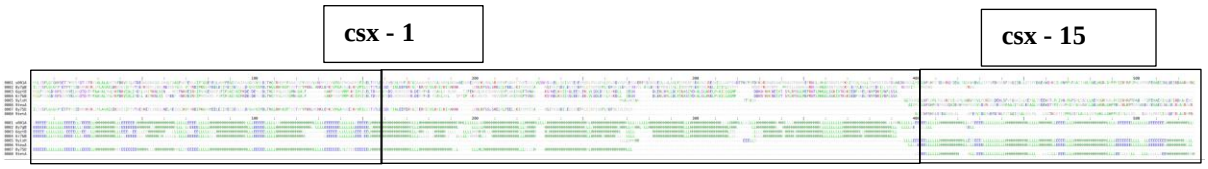


Figure 9 Overall, from DALI's hit

3.4 Domain prediction:

The CHAINSAW tool predicted that baltymas 1 have three domains:

Domain1 : 2 – 158

Domain 2: 159 – 387

Domain 3: 401 – 525

Where the first and second domains are CARF domain (N-terminal), HEPN domain (middle) that is also known as csx1 and C-terminal domain is csx-15.

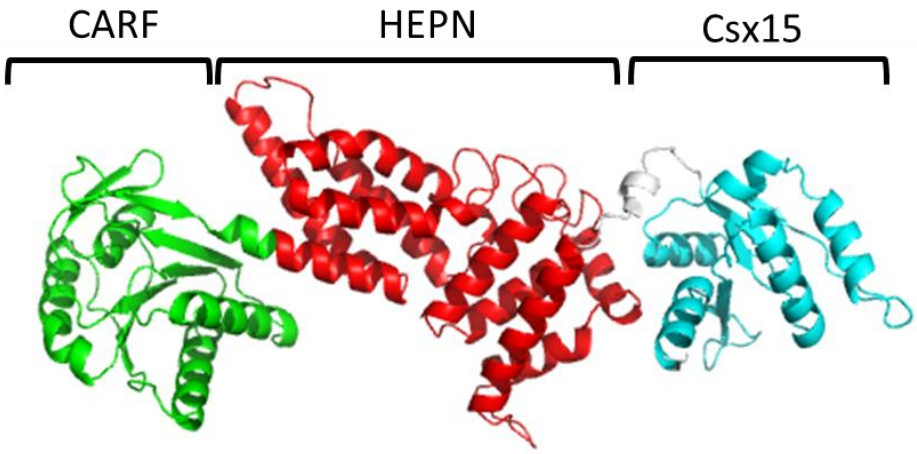


Figure 10 Domain prediction for Baltymas1 given by Chainsaw. Different colours in structure highlight domains predicted by the tool

Supplementary:

Structural_bioinformatics_files > Supplementary > chainsaw > baltymas1.pse / baltymas_1.tsv

3.5 Baltymas_1 existence state

Alpha fold 3 helped to analyze the stable state of baltymas 1 by generating the iPTM score for the protein which confirms if it is in monomeric, dimeric, trimeric, or quaternary structure. The modelling suggested that the fused Csx 1 – Csx 15 construct preferably stable or adopts a monomeric state but the structure quality is better in tetramer with 0.589 VoroMQA score (**Table 1**). This suggests that although baltymas 1 is likely monomeric, it may adopt a tetramer form and potentially perform its function more effectively in this state.

Table 1:

S.No	Types of model	IPTM score	PTM score	VoroMQA score
1.	Baltymas 1 monomer	--	0.71	0.519
2.	Baltymas 1 dimer	0.64	0.66	0.563
3.	Baltymas 1 trimer	0.37	0.48	0.562
4.	Baltymas 1 tetramer	0.33	0.43	0.589
5.	Baltymas 1 tetramer-RNA	0.33	0.43	0.584

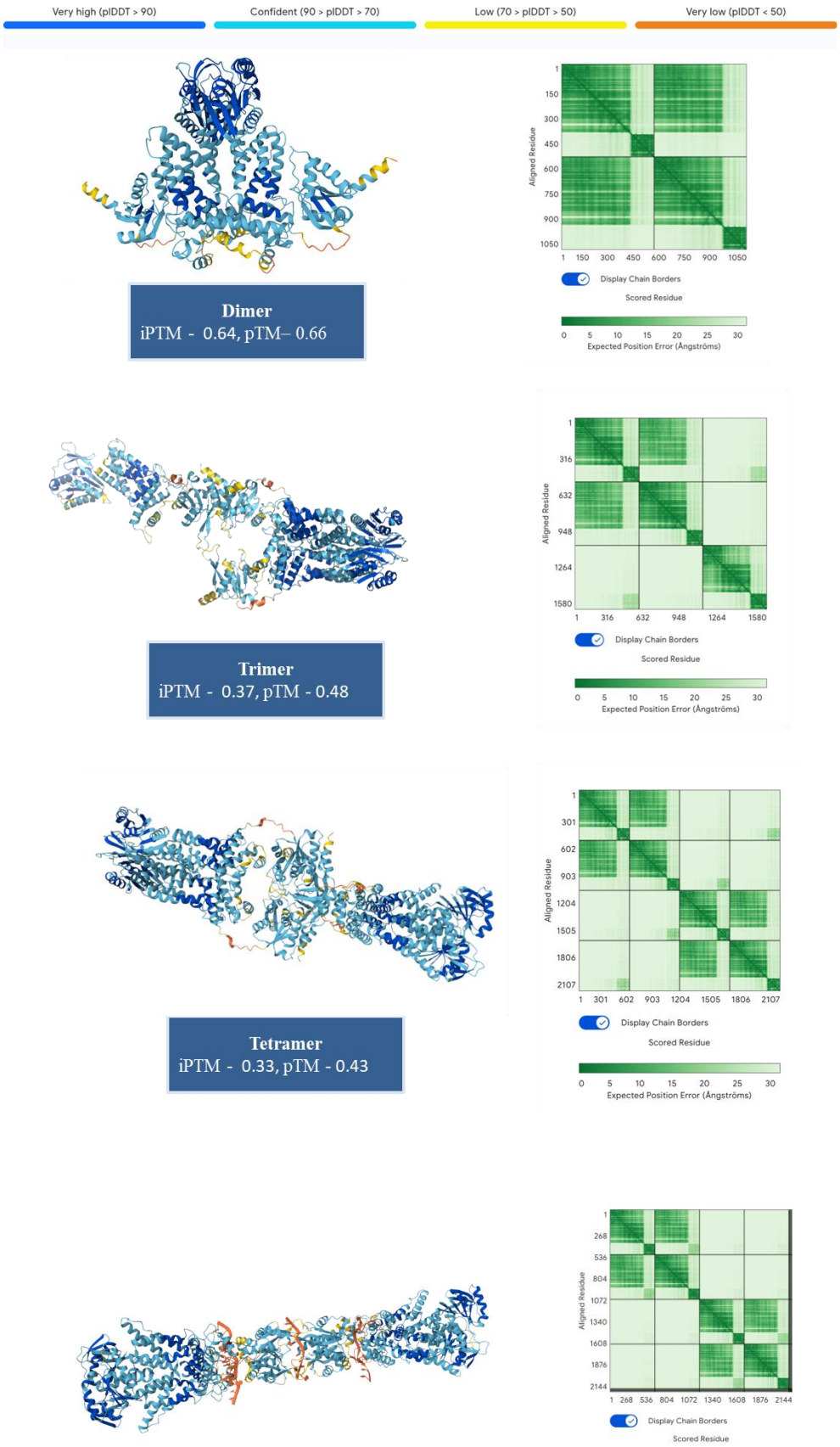


Figure 11 Protein modelling of Baltymas 1 in different states with Alphafold 3

3.6 Evolutionary conserved sites of Baltymas_1

The Baltymas 1 shows the highly conserved residues located in active sites, suggesting functional importance. But for the 3rd domain ConSurf showed an insufficient data.

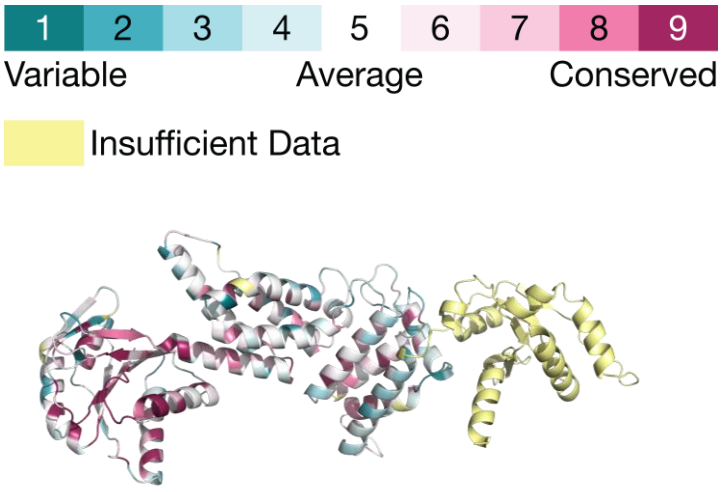


Figure 12 ConSurf analysis of baltymas 1

Supplementary:

Structural_bioinformatics_files > Supplementary > ConSurf > baltymas1_consurf_pymol_session.pse

3.7 Enzymatic activity and heteromeric interaction with protein

The predicted domain architecture of Baltymas 1 revealed a CRISPR-associated Csx1 protein. While both Csx1 and Csx15 have been reported or predicted to form stable dimers individually.

Csx1 have two domains CARF and HEPN. CARF domain predicted to have active sites Lysine-12, Tyrosine-15, Tyrosine-16, Arginine-132, Glutamate-59, Tyrosine164 which plays a major role in cA_4 binding and signalling [12]. Structural analysis by PyMOL revealed that 1st domain of baltymas 1 exhibits a similar fold and conservation of residues at Threonine-9, Tyrosine-12, serine-13, arginine-111, glutamate- 49, arginine- 142 positions. This suggests that N-terminal domain of baltymas 1 may perform CARF-like function. Shown in figure 13

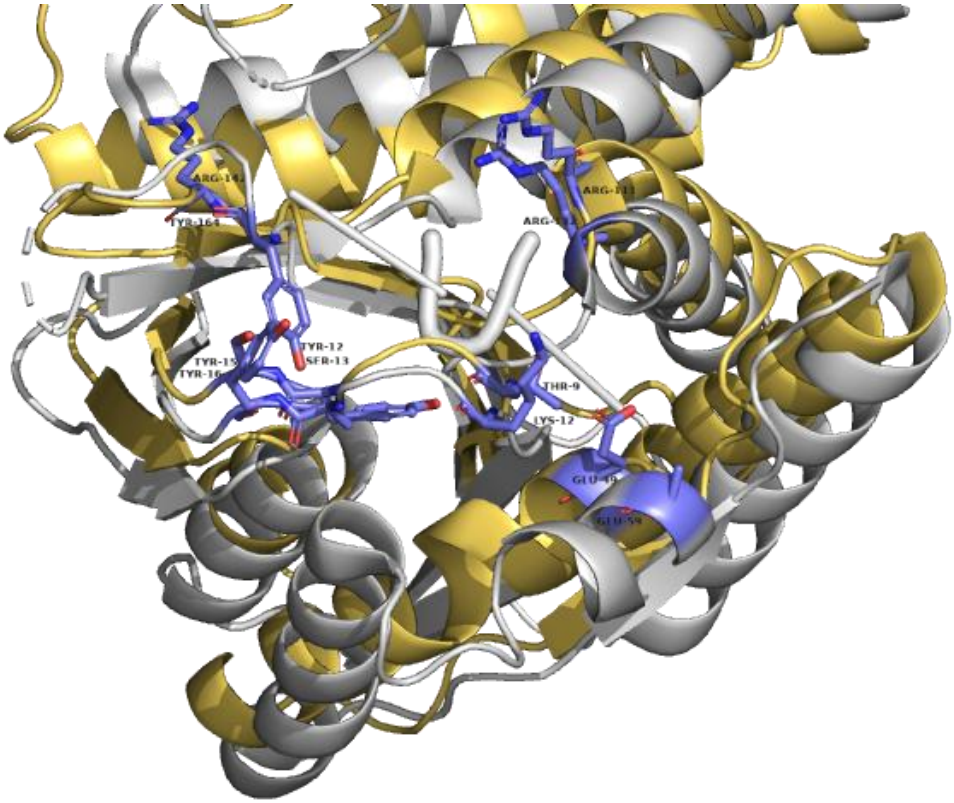


Figure 13 CARF domain of Baltymas 1 where yellow color- baltymas1 and grey color – superimposed with csx1 CARF domain.

On the other hand, HEPN is proposed to bind and cleave single-stranded RNA (ssRNA) in a relatively non-specific manner, with the help of active sites present at Arginine-401, Histidine-406, Arginine-345, Asparagine-402, Glutamate-369 positions. [13] the 2nd domain of baltymas 1 also shows the presence of comparable sites at arginine-4.1, histadine-358, arginine-307, Aspargine-354, lysine-337 shown in figure 14. Therefore, it is possible that this domain may possess HEPN-like role.

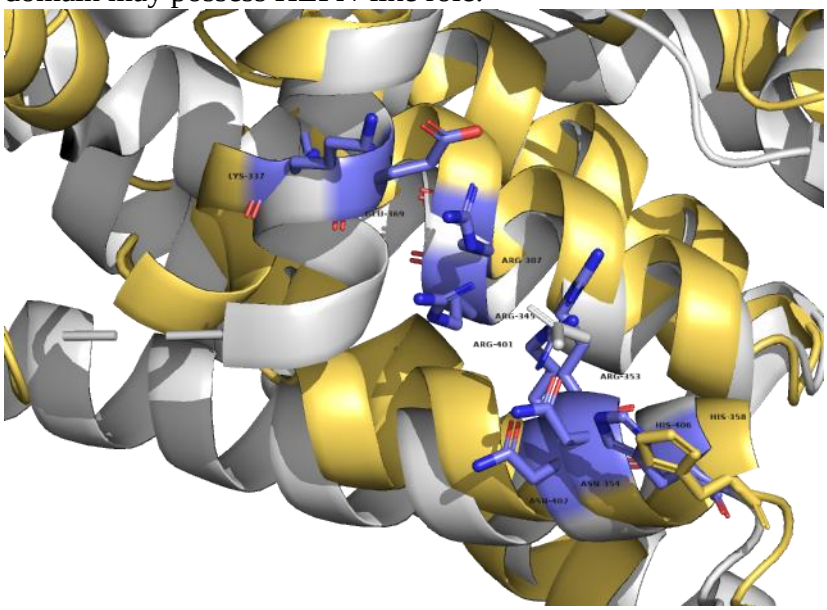


Figure 14 HEPN domain of Baltymas 1 color indication- yellow color- Baltymas1 2nd domain and grey color- superimposed with HEPN domain of csx1

Csx 15 shows enzymatic activity through its architecture in which multi-stack head-to-tail, generating a shared binding site formed at the dimer:dimer interface, where cA4 molecules can bind. Cyclic tetra-adenylate (cA_4) is a crucial second messenger molecule in prokaryotes, synthesized by Type III CRISPR-Cas immune systems upon detecting invading viral (phage) RNA. [14] Followed by the ConSurf results and superimpose, the conserved residues like Histadine-13, Arginine-123, and Histadine-90 coordinate the phosphate groups and adenosine bases of cA4.[14] the C-terminal of baltymas 1 also have similar fold or residues at comparable position that is- His-408, PHE-507/GLU-508, ARG-486So, this also proves that it is likely to have heteromeric protein-protein interactions. Shown in figure 15

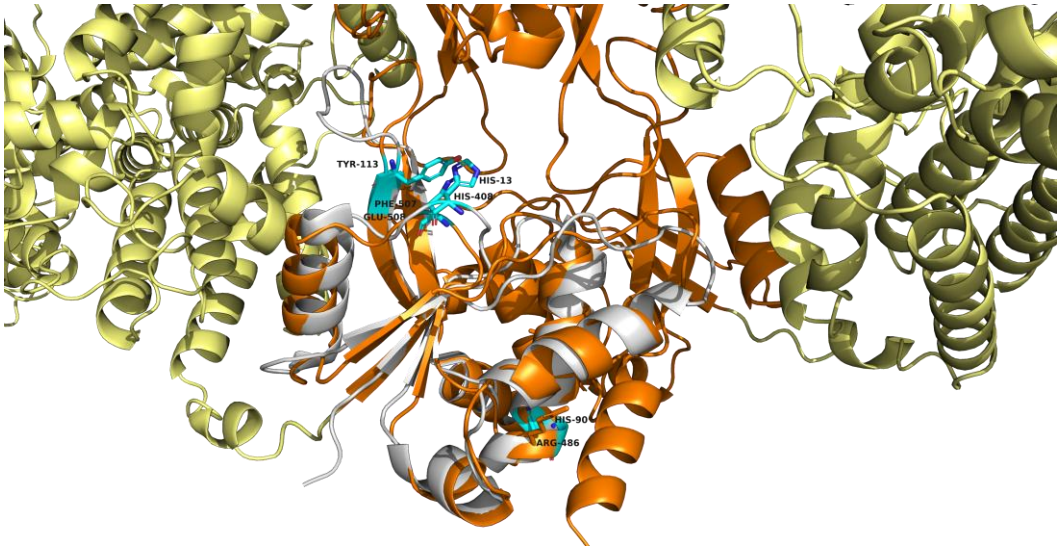


Figure 15 active sites of baltymas_1 3rd domain by superimposing with Csx15, the color indication is yellow= tetramer of Csx1 and orange = dimer:dimer of domain3, grey = csx15 superimposed with one of the 3rd domain of baltymas1

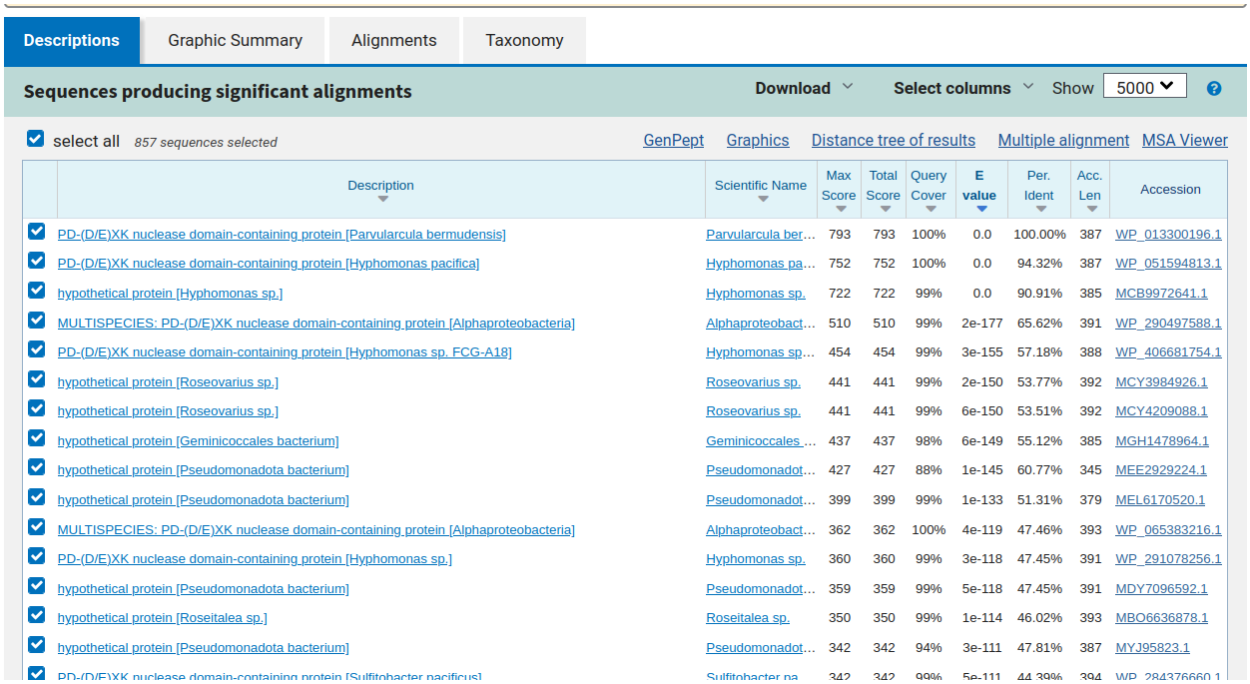
Supplementary: Structural_bioinformatics_files > Supplementary > DALI > baltymas1_csx1_active_site_pymol

4. Results of Baltymas 2

4.1 Sequence analysis:

The first step involved identifying homologous proteins of Baltymas 2. The query protein was searched against non-redundant database of BLASTp, PSI-BLAST and uniRef50. The BLASTp results indicate that baltymas 2 is a homolog to the **PD-(D/E)XK nuclease domain-containing protein (figure 14)**. Because of these reasons:

- E-Value = 0.0
- 99-100% query coverage
- 90-100% sequence identity for closest homologs



Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
PD-(D/E)XK nuclease domain-containing protein [Parvularcula bermudensis]	Parvularcula ber...	793	793	100%	0.0	100.00%	387	WP_013300196.1
PD-(D/E)XK nuclease domain-containing protein [Hyphomonas pacifica]	Hyphomonas pa...	752	752	100%	0.0	94.32%	387	WP_051594813.1
hypothetical protein [Hyphomonas sp.]	Hyphomonas sp.	722	722	99%	0.0	90.91%	385	MCB9972641.1
MULTISPECIES: PD-(D/E)XK nuclease domain-containing protein [Alphaproteobacteria]	Alphaproteobact...	510	510	99%	2e-177	65.62%	391	WP_290497588.1
PD-(D/E)XK nuclease domain-containing protein [Hyphomonas sp. FCG-A18]	Hyphomonas sp...	454	454	99%	3e-155	57.18%	388	WP_406681754.1
hypothetical protein [Roseovarius sp.]	Roseovarius sp.	441	441	99%	2e-150	53.77%	392	MCY3984926.1
hypothetical protein [Roseovarius sp.]	Roseovarius sp.	441	441	99%	6e-150	53.51%	392	MCY4209088.1
hypothetical protein [Geminicoccales bacterium]	Geminicoccales ...	437	437	98%	6e-149	55.12%	385	MGH1478964.1
hypothetical protein [Pseudomonadota bacterium]	Pseudomonadot...	427	427	88%	1e-145	60.77%	345	MEE2929224.1
hypothetical protein [Pseudomonadota bacterium]	Pseudomonadot...	399	399	99%	1e-133	51.31%	379	MEL6170520.1
MULTISPECIES: PD-(D/E)XK nuclease domain-containing protein [Alphaproteobacteria]	Alphaproteobact...	362	362	100%	4e-119	47.46%	393	WP_065383216.1
PD-(D/E)XK nuclease domain-containing protein [Hyphomonas sp.]	Hyphomonas sp.	360	360	99%	3e-118	47.45%	391	WP_291078256.1
hypothetical protein [Pseudomonadota bacterium]	Pseudomonadot...	359	359	99%	5e-118	47.45%	391	MDY7096592.1
hypothetical protein [Roseitalea sp.]	Roseitalea sp.	350	350	99%	1e-114	46.02%	393	MBO6636878.1
hypothetical protein [Pseudomonadota bacterium]	Pseudomonadot...	342	342	94%	3e-111	47.81%	387	MYJ95823.1
PD-(D/E)XK nuclease domain-containing protein [Sulfitobacter pacificus]	Sulfitobacter pa...	342	342	99%	5e-111	44.39%	394	WP_284376660.1

Figure 16 BLASTp search against non-redundant databases from NCBI for Baltymas_2

The HHpred analyses have suggested that the C-terminal of this Baltymas_2 contains a conserved domain. It also shows the heterogeneous hits with Dpnull, DUF8371, PD-(D/E) XK, RE_Banl_C, Sugar ferm, and many more. Many of these belongs to PD-(D/E)XK nuclease superfamily. Whereas N-terminal region (1-223 appears to have few or no strong matches). Which means it could be a novel domain or highly diverged domain or domain not represented well in current databases (figure 15).

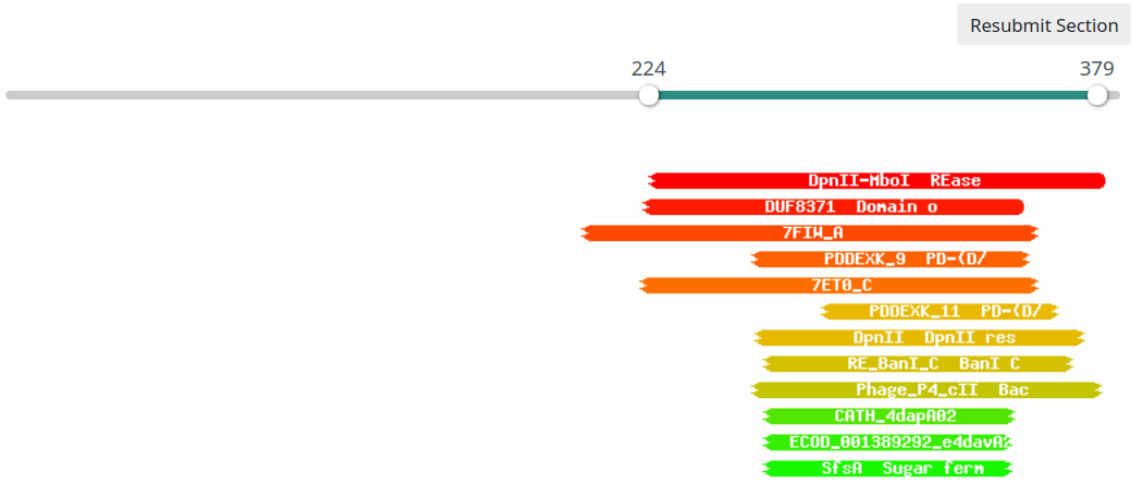


Figure 17 HHpred results for baltymas_2

4.2 Structure prediction:

The 3D structure of the baltymas 2 sequence was analysed and generated by using the alphafold3 server as shown in (Figure 18). The iPTM scores of 0.6 indicated moderate confidence in the intermolecular interface. The 3D structure shows the pLDDT higher values, and few terminals displayed the lower confidence scores. This suggests that the overall predicted structure closely resembles its true confirmation.

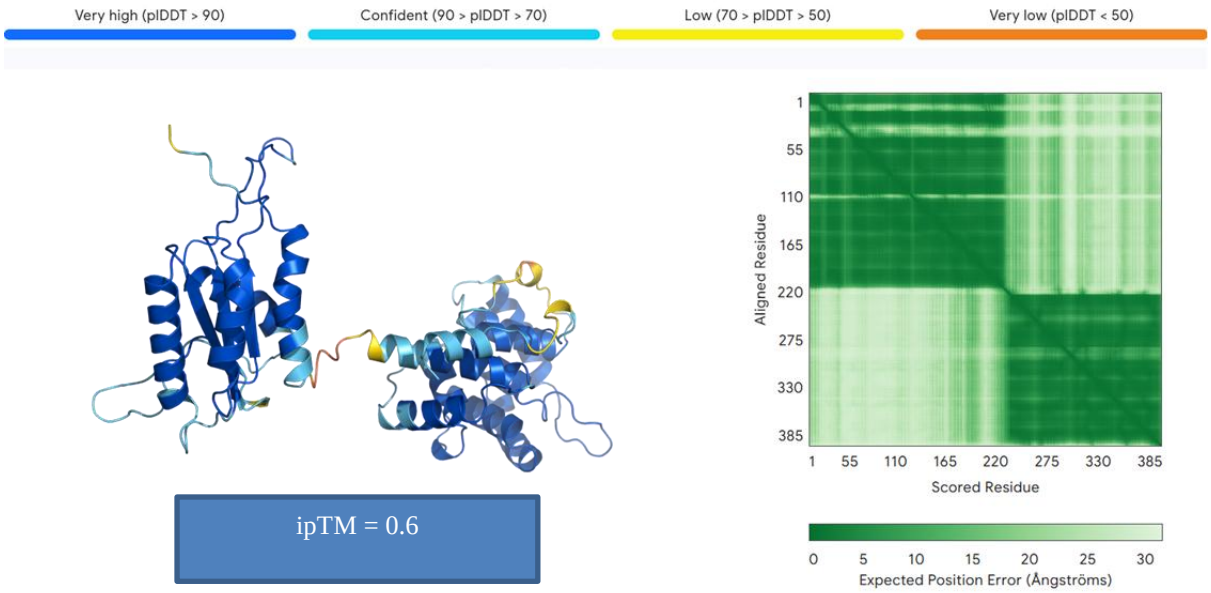
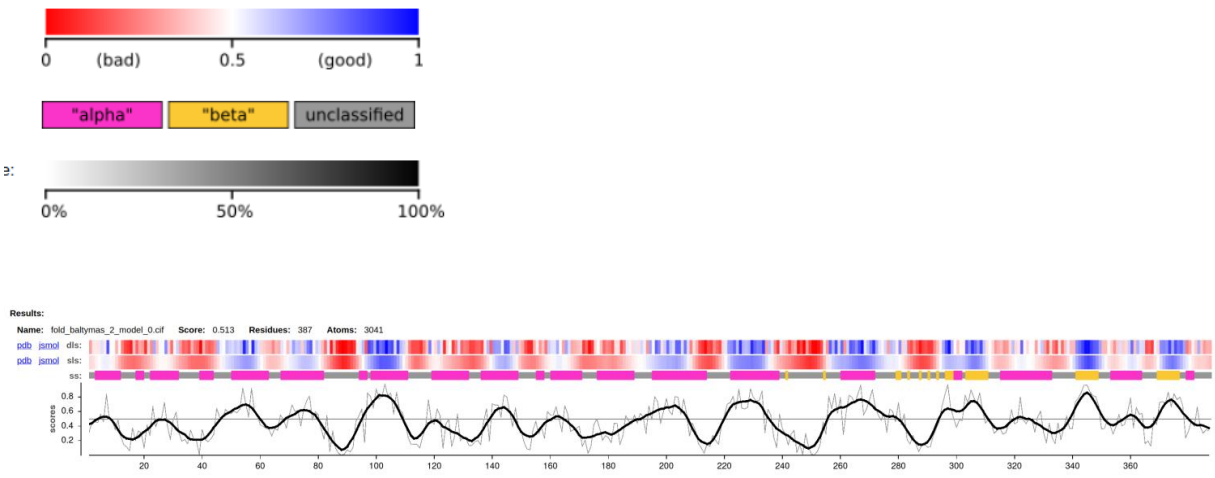


Figure 18 Baltymas 2 prediction from alphafold 3 server (https://alphafoldserver.com/) and the 3D structure is visulaised by PyMOL.

The structure quality analysis of baltymas 1 was done by VoroMQA. The scores from VoroMQA were reported to be 0.519 as shown in figure 19, indicating overall good quality.



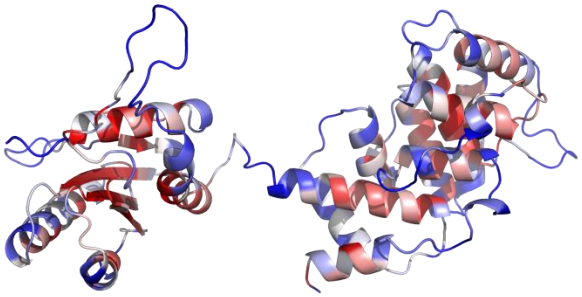


Figure 19 VoroMQA analysis of Baltymas 2 for structure quality analysis. And the generated structure visulaised with the help of PyMOL

4.3 Domain segmentation

Then DALI tool was used that matches against PDB90 (interactive HTML) to analyse the unknown domain from the 3D protein structure.

Results: baltymas2

Query: s001A

Select neighbours (check boxes) for viewing as multiple structural alignment or 3D superimposition. The list of neighbours is sorted by Z-score. Similarities with a Z-score lower than 2 are spurious. Each neighbour has links to pairwise structural alignment with the query structure, and to the PDB format coordinate file where the neighbour is superimposed onto the query structure.

Structural Alignment | Expand gaps | 3D Superimposition (PV) | SANS | PANZ | Pfam | Reset Selection

Summary

The list is truncated at 100 hits.

No:	Chain	Z	rmsd	lali	nres	%id	PDB	Description
1:	4xuv-A	10.2	3.9	194	332	8	PDB	MOLECULE: GLYCOSIDE HYDROLASE FAMILY 105 PROTEIN;
2:	5noa-A	9.9	4.2	194	354	4	PDB	MOLECULE: FAMILY 88 GLYCOSYL HYDROLASE;
3:	3qwt-A	9.6	3.9	193	378	6	PDB	MOLECULE: PUTATIVE GH105 FAMILY PROTEIN;
4:	3pmm-A	9.5	3.9	191	381	5	PDB	MOLECULE: PUTATIVE CYTOPLASMIC PROTEIN;
5:	3k11-A	9.3	5.5	209	439	5	PDB	MOLECULE: PUTATIVE GLYCOSYL HYDROLASE;
6:	4q88-A	9.1	4.0	183	350	4	PDB	MOLECULE: UNCHARACTERIZED PROTEIN;
7:	8s3o-A	9.1	15.0	189	1324	7	PDB	MOLECULE: CD109 ANTIGEN;
8:	5bp8-A	9.0	4.4	175	515	9	PDB	MOLECULE: ENT-COPALYL DIPHOSPHATE SYNTHASE;
9:	3k7x-A	8.7	4.0	195	348	7	PDB	MOLECULE: LIN0763 PROTEIN;
10:	2sqc-A	8.6	4.0	180	623	9	PDB	MOLECULE: SQUALENE-HOPENE CYCLASE;
11:	8paj-A	8.6	4.0	186	342	7	PDB	MOLECULE: SQUALENE-HOPENE/TETRAPRENYL-BETA-CURCUMENE CYCLAS
12:	8ixt-A	8.6	11.9	190	401	9	PDB	MOLECULE: TRANSCOBALAMIN-2;
13:	7fiv-B	8.5	8.1	139	741	6	PDB	MOLECULE: CIDA_I GAMMA/2 PROTEIN;
14:	6shm-A	8.5	4.1	185	339	11	PDB	MOLECULE: MUTANT ALPHA-1,6-MANNANASE GH76A;
15:	2bb6-A	8.5	11.0	184	414	11	PDB	MOLECULE: TRANSCOBALAMIN II;
16:	4kki-A	8.5	4.0	174	398	5	PDB	MOLECULE: TRANSCOBALAMIN-1;
17:	6sbc-A	8.3	4.2	182	352	10	PDB	MOLECULE: MSTE;
18:	6vpt-A	8.3	3.8	171	483	7	PDB	MOLECULE: CYCLASE;
19:	4boj-C	8.3	4.2	186	344	7	PDB	MOLECULE: ALPHA-1,6-MANNANASE;
20:	7bjt-A	8.1	16.4	199	727	8	PDB	MOLECULE: ALGINATE LYASE, FAMILY PL17;
21:	1w6j-A	8.1	4.1	185	727	5	PDB	MOLECULE: LANOSTEROL SYNTHASE;
22:	1nc5-A	8.1	4.2	184	363	6	PDB	MOLECULE: HYPOTHETICAL PROTEIN YTER;
23:	4v1r-A	8.1	4.1	193	361	4	PDB	MOLECULE: ALPHA-1,6-MANNANASE;
24:	2bb5-A	8.1	14.7	191	409	10	PDB	MOLECULE: TRANSCOBALAMIN II;
25:	4ce7-B	8.1	4.0	184	348	3	PDB	MOLECULE: UNSATURATED 3S-RHAMNOGLYCURONYL HYDROLASE;
26:	8oak-B	8.0	4.2	183	347	8	PDB	MOLECULE: SQUALENE--HOPENE CYCLASE:

Figure 20 DALI results for Baltymas 2

As the results from DALI was retrieved that indicates the 10.2 Z-scores indicates good structural similarity, 3.9 RMSD, which consider as good because the number of amino acid residues (nres) that are included in the structural alignment between the query structure and the first hit of DALI structure is 332.

To further evaluate the structural similarity, DALI's first hit (4xuv-A) was super imposed with baltymas 2 (as shown in figure 21). Which was visualised by PyMOL. However, as the

initial full-length DALI search yielded no significant structural similarity, it was hypothesized that the protein may comprise multiple structurally independent domains that could obscure the overall similarity when analyzed as a whole.

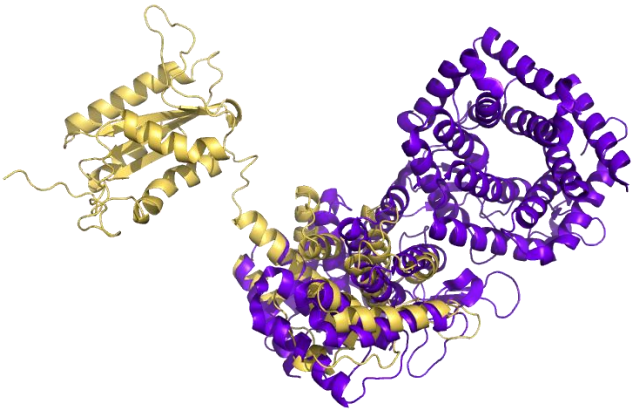


Figure 21 This is the visulization from PyMOL which shows the super impose of baltymas 2(yellow) and DALI's first hit (purple 4xuv) 3-D structure.

Therefore, domain segmentation was carried out using the CHAINSAW tool for Baltymas_2. It predicted that the query protein has two functional domains as shown in figure 22:

- Domain 1: 3-213
- Domain 2: 222 – 381

The domain chopping was performed with a high confidence score of 0.961, indicating reliable domain boundary prediction.

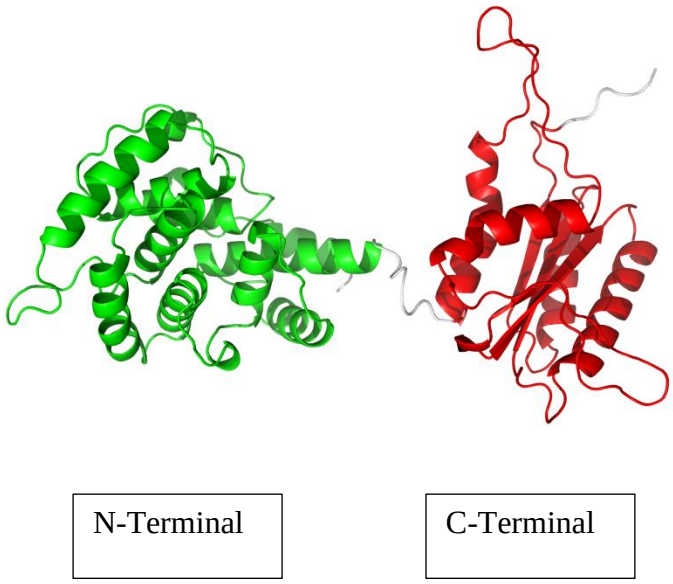


Figure 22 CHAINSAW domain prediction of Baltymas 2, viewed on PyMOL

Supplementary:

Structural_bioinformatics_files > Supplementary > chainsaw > baltymas_2.pse / baltymas_2.tsv

To further investigate the structural similarity of the individual domains of Baltymas_2 each domain was superimposed to its closest structural homolog identified through DALI search. The N-terminal domain (Domain 1, residues 3–213) was super imposed with 4ce7-B (the unsaturated 3S- Rhamnoglycuronyl hydrolyase) as shown in figure 23. It has Z-score of 9.6 and rmsd of 4 with 192 – 348 aligned residues (as shown in figure 24).

No:	Chain	Z	rmsd	lali	nres	%id	PDB	Description
1:	4xuv-A	10.1	3.9	193	332	8	PDB	MOLECULE: GLYCOSIDE HYDROLASE FAMILY 105 PROTEIN;
2:	4ce7-B	9.6	4.0	192	348	5	PDB	MOLECULE: UNSATURATED 3S-RHAMNOGLYCURONYL HYDROLASE;
3:	3qwt-A	9.6	3.9	192	378	6	PDB	MOLECULE: PUTATIVE GH105 FAMILY PROTEIN;
4:	3pmm-A	9.5	3.8	191	381	6	PDB	MOLECULE: PUTATIVE CYTOPLASMIC PROTEIN;
5:	3k11-A	9.4	4.3	204	439	4	PDB	MOLECULE: PUTATIVE GLYCOSYL HYDROLASE;
6:	5noa-A	9.3	4.2	189	354	4	PDB	MOLECULE: FAMILY 88 GLYCOSYL HYDROLASE;
7:	4q88-A	9.2	4.0	186	350	4	PDB	MOLECULE: UNCHARACTERIZED PROTEIN;
8:	5bp8-A	9.2	4.5	174	515	9	PDB	MOLECULE: ENT-COPALYL DIPHOSPHATE SYNTHASE;
9:	8s3o-A	9.1	4.3	176	1324	7	PDB	MOLECULE: CD109 ANTIGEN;
10:	8ixt-A	9.1	4.2	188	401	10	PDB	MOLECULE: TRANSCOBALAMIN-2;
11:	7bjt-A	9.0	3.5	169	727	9	PDB	MOLECULE: ALGINATE LYASE, FAMILY PL17;
12:	8paj-A	8.7	4.2	186	342	8	PDB	MOLECULE: SQUALENE-HOPENE/TETRAPRENYL-BETA-CURCUMENE CYCLAS
13:	4ojz-A	8.6	3.3	169	706	9	PDB	MOLECULE: PUTATIVE ALGINATE LYASE;
14:	5i5k-B	8.6	4.3	189	1634	6	PDB	MOLECULE: COMPLEMENT C5;
15:	2sqc-A	8.4	3.9	177	623	8	PDB	MOLECULE: SQUALENE-HOPENE CYCLASE;

Hydrolase protein

Figure 23 DALI server search results for Baltymas 2 Domain 1 (N-terminal).

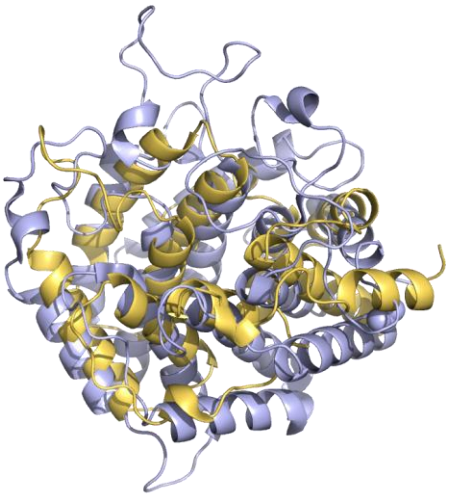


Figure 24 Structural superimposition of Baltymas 2 Domain 1 (N-terminal) with 4ce7 – B (DALI's second hit)

Similarly, the C-terminal domain was superimposed with 4g6u – A (EC869 CDIA – CT) as shown in figure 25, it had Z-scores of 8.4, rmsd 2.8 with 113 – 219 aligned residues. Where it superimposes with the C-terminal of CdiA-CT/CdiI toxin and immunity complex from Escherichia coli and it exist in monomeric state (figure 26). [14]

No:	Chain	Z	rmsd	lali	nres	%id	PDB	Description
1:	7fiv-B	8.5	2.6	126	741	6	PDB	MOLECULE: CIDA_I GAMMA/2 PROTEIN;
2:	4g6u-A	8.4	2.8	113	213	12	PDB	MOLECULE: EC869 CDIA-CT;
3:	1y88-A	7.6	3.7	108	184	13	PDB	MOLECULE: HYPOTHETICAL PROTEIN AF1548;
4:	4zqu-A	7.5	2.9	102	124	9	PDB	MOLECULE: CDIA-CT TOXIN, CONSERVED DOMAIN PROTEIN;
5:	9ibq-A	7.4	4.6	129	286	13	PDB	MOLECULE: BSTEII;
6:	1d2i-A	7.3	5.2	127	223	12	PDB	MOLECULE: DNA (5'-D(*TP*AP*TP*TP*AP*TP*AP*GP*AP*TP*CP*TP*AP
7:	1esg-B	7.2	3.7	128	212	11	PDB	MOLECULE: DNA (5'-D(*TP*GP*GP*AP*TP*TP*CP*A)-3');
8:	3odh-A	7.1	3.8	123	194	7	PDB	MOLECULE: OKRAI ENDONUCLEASE;
9:	6xl1-A	6.8	2.9	111	373	9	PDB	MOLECULE: CAD1;
10:	7vsr-M	6.8	3.9	122	296	12	PDB	MOLECULE: 5-METHYLCYTOSINE-SPECIFIC RESTRICTION ENZYME B;
11:	1vrr-A	6.7	3.5	121	203	10	PDB	MOLECULE: 5'-D(*TP*TP*AP*TP*AP*GP*AP*TP*CP*TP*AP*TP*AP*A)-
12:	8ps5-A	6.6	3.1	113	408	12	PDB	MOLECULE: SHEDU EFFECTOR PROTEIN;
13:	5gke-A	6.6	5.4	97	239	15	PDB	MOLECULE: ENDONUCLEASE ENDOMS;
14:	3bm3-A	6.6	4.8	118	259	14	PDB	MOLECULE: DNA (5'-D(*CP*AP*TP*CP*CP*AP*GP*GP*TP*AP*C)-3');
15:	2inb-A	6.5	3.0	106	128	10	PDB	MOLECULE: HYPOTHETICAL PROTEIN;

Figure 25 DALI server search results for Baltymas 2 Domain 2 (C-terminal).

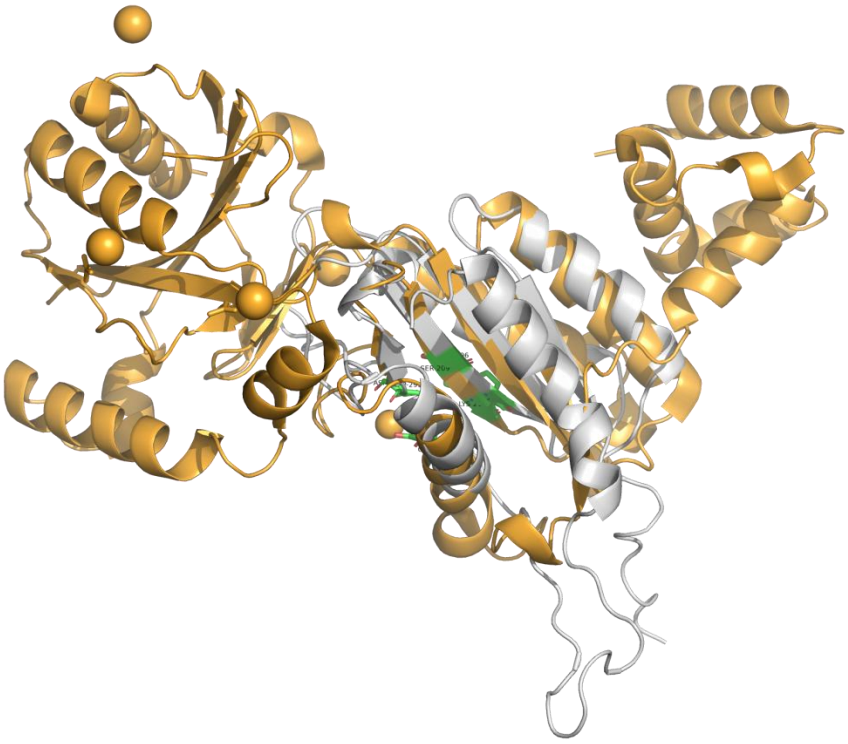


Figure 26 Structural superimposition of Baltymas 2 Domain 2 (C-terminal) with 4g6u – A (DALI's hit)

The data so far suggests that Baltymas_2 has two domains, where C-terminal of baltymas 2 shows the homology with C-terminal of Ecoli toxin protein that is a α/β nuclease domain, bound to its specific immunity protein and N-terminal domain belongs to glycoside hydrolase family

4.4 Baltymas_2 existence state

Alpha fold 3 helped to analyze by generating the iPTM score for the protein which confirms if it is in monomeric, dimeric, trimeric, or quaternary structure. The results provide strong suggestions that Baltymas_2 interacts with $Zn2^{+}$/ $Mg2^{+}$, with the models achieving highly promising pTM+ipTM score (~0.82-0.95) as shown in (**Table 2**). Based on structural prediction confidence metrics, the C-terminal dimer bound to $Zn2^{+}$/ $Mg2^{+}$ appears to be the most reliable/well-predicted assembly state. However, Baltymas2 – Zn2+ monomer and Baltymas2– Mg2+ monomer shows to be reliable state.

Table-2

S.No	Type of Models	iPTM score	pTM score	VoroMQA score
1.	Baltymas 2 monomer	-	0.6	0.513
2.	Baltymas 2 dimer	0.45	0.47	0.550
3.	Baltymas 2 trimer	0.2	0.3	0.529
4.	Baltymas 2 te0tramer	0.29	0.35	0.522
5.	Baltymas_2 monomer – Zn2+	0.9	0.54	0.513
6.	Baltymas_2– $Mg2^{+}$ monomer	0.91	0.55	0.551
7.	Baltymas_2 C-terminal– Zn2+ monomer	0.95	0.86	0.565
8.	Baltymas2 C-terminal– $Mg2^{+}$ monomer	0.91	0.89	0.553
9.	Baltymas_2 – $Mg2^{+}$ di-mer	0.42	0.46	0.539
10.	Baltymas_2 – $Zn2^{+}$ di-mer	0.48	0.51	0.553
11.	Baltymas_2-C-terminal $Mg2^{+}$ dimer	0.82	0.87	0.606
12.	Baltymas_2-C-terminal $Zn2^{+}$ dimer	0.85	0.87	0.610
13.	Baltymas_2 – $Mg2^{+}$ tri-mer	0.26	0.35	0.524
14.	Baltymas_2 – $Zn2^{+}$ tri-mer	0.29	0.36	0.531
15.	Baltymas_2-C-terminal $Mg2^{+}$ trimer	0.26	0.35	0.524
16.	Baltymas_2-C-terminal $Zn2^{+}$ trimer	0.48	0.62	0.591

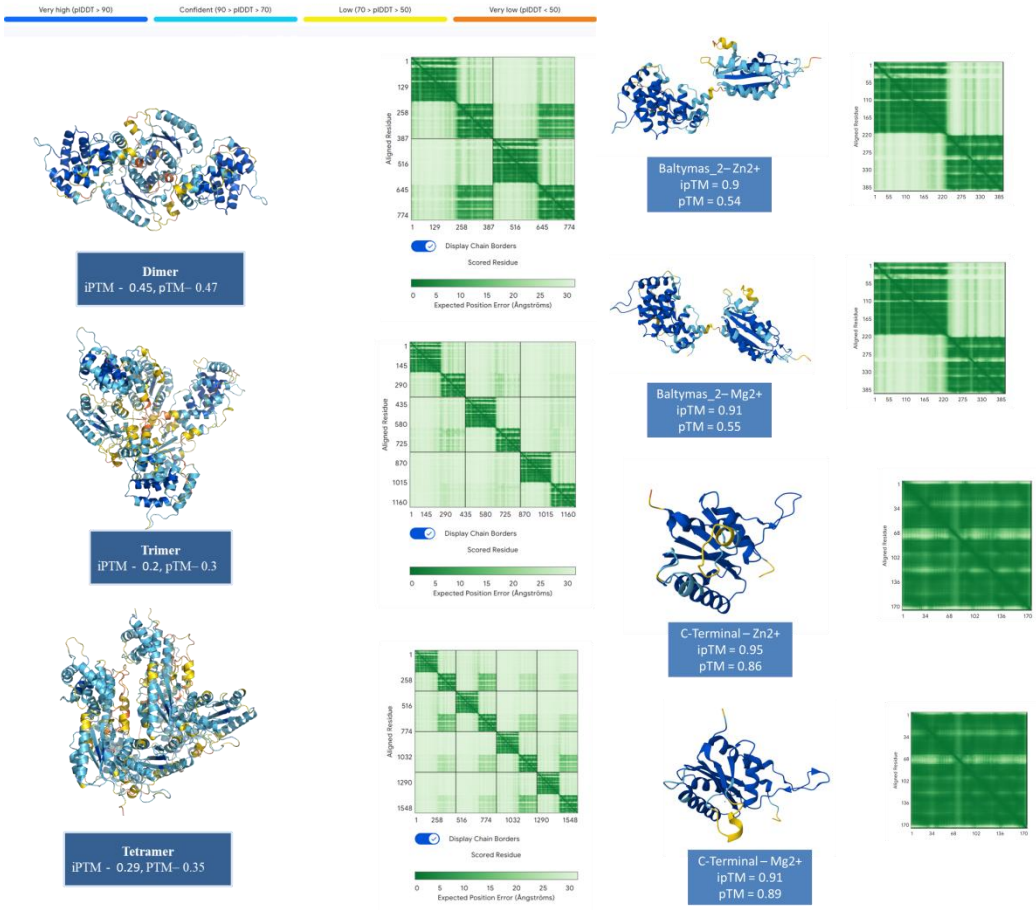


Figure 27 Protein modelling of Baltymas 2 in different states with Alphafold 3

4.5 Evolutionary conserved sites of Baltymas_2

ConSurf analysis of baltymas 2 showed the contrasting patterns of evolutionary conservation. The N-terminal region displayed the variable to average conservation (scores ~ 1-6), which means it has greater sequence variability. It implies that this region may or may not have hydrolase activity.

On the other hand, the C terminal region demonstrate high conservation scores (~ 8-9), indicating strong evolutionary conservation of this domain. This suggests that C-terminal domain carries functionally active residues that are essential for the E.coli toxin protein's biological activity and etc.

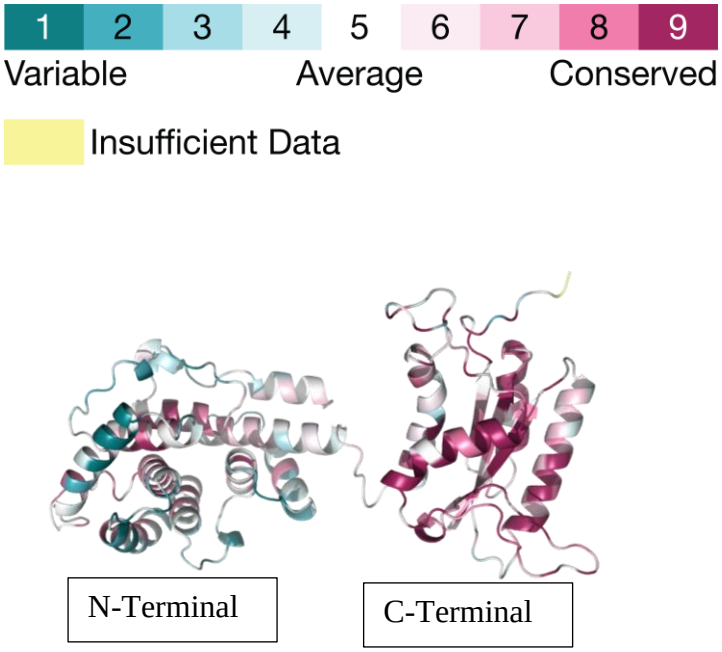


Figure 28 ConSurf analysis of baltymas 2

Supplementary:

Structural_bioinformatics_files > Supplementary > ConSurf > baltymas_2_consurf_pymol_session.pse

4.6 Active site and enzymatic activity

C-terminal Baltymas_2

The analysis revealed so far that the catalytic residues LYS211, and ASP198 are conserved in 4g6u protein and seems to have same active region in the query protein's domain –LYS308 and ASP 295, this suggest that they both may share the enzymatic mechanism as shown in figure 28. However, GLU 177 and SER 209 from 4g6u is substituted by GLN 264 and GLU-306 query protein's domain. This recommends that baltymas 2 C-terminal domain might function as DNase if it combines with Mg2+ or Zn2+. Moreover, In this toxin, **Glu177** and **Asp198** are two amino acids that help hold the metal ion, so the toxin can cut DNA. [15]

Whereas, the N-terminal predicted to have hydrolase activity with its residues but still, it is difficult to imagine the arrangement of this model in the membrane without having sufficient data.

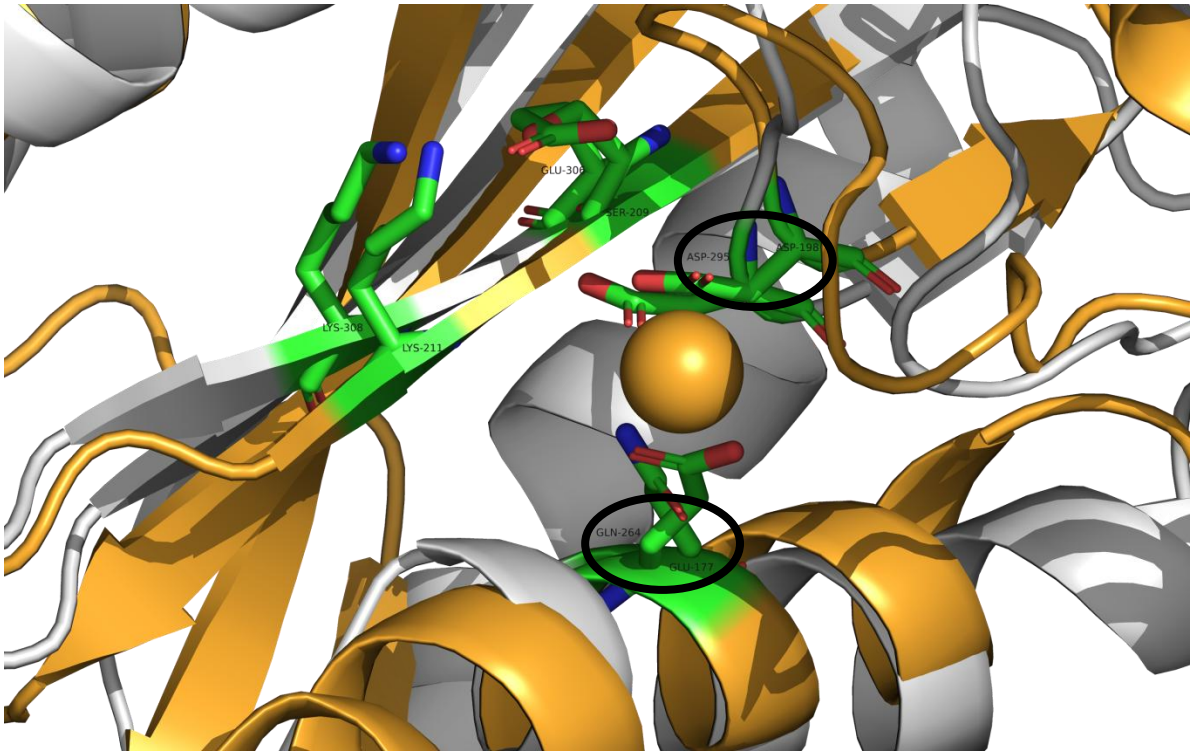


Figure 29 Active sites of C-terminal of baltymas 2 compared with DALI's 4g6u hit

Supplementary:

Structural_bioinformatics_files > Supplementary > DALI > baltymas2-C-Terminal_active_sites

5. Conclusion

Baltymas 1

In conclusion of baltymas 1, it belongs to CRISPR -associated proteins and have three domains which could have potential to perform functionally uniquely. 1st domain might have ability in cA_4 binding and signalling, 2nd domain may bind and cleave ssRNA in a relatively non-specific manner and 3rd domain could perform herteromeric activity since it has the binding sites between dimer:dimer interface**.** may play a transport-related role in cellular processes within the CRISPR-Cas system

Baltymas 2

While its exact function remains unclear, the interaction with ions suggests that C-terminal domain of baltymas2 alone exist in dimeric form and may perform DNase, since it have the similar fold and residues those are responsible for DNase activity in the protein. But the activity of this domain is also dependent on the protein is is binding too. The N-terminal region of Baltymas_2 showed similarity to members of the hydrolase protein family. However, due to the lack of published experimental data, its enzymatic activity and interaction mechanisms cannot be confirmed at present.

6. Acknowledgement

I would like to thank Dr. Ceslovas Venclovas for his guidance and support throughout the semester. I am greatful for the oppourtunity to learn and develop my skills in structural bioinformatics.

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